Week	Dates	Textbook Ref.	Weekly Content Topics	Even Lab Sections	Odd Lab Sections
Week 1	Aug 31 - Sep 4	OSv1: 15.1, 16.5 EN1	Introduction; simple harmonic motion (SHM) of systems obeying Hooke's law; differential equation of SHM; superposition of SHMs of the same frequency.	No Labs	
Week 2	Sep 7 (holiday) Sep 8 - Sep 11	OSv1: 16.2 EN1, EN2	Phasors; phasor addition; phasor diagrams; phasors as complex quantities; traveling wave pulse: mathematical description and particle motion; partial derivatives; harmonic wave equation.	No Labs	
Week 3	Sep 14 - Sep 18	OSv1: 16.5, 17.1, 17.2 EN2, EN3	Principle of wave superposition; harmonic (sinusoidal) waves; mathematical description, particle motion, phase differences; complex representation of a traveling harmonic wave; speed of waves in real media: stretched string; sound waves as longitudinal waves in solids, liquids, and gases; speed of sound waves.		Lab 1
Week 4	Sep 21 - Sep 25	OSv1: 15.2, 16.4 EN4	Energy transport by a harmonic wave; mechanical impedance; power transported by a wave; acoustical impedance.	Lab 1	
Week 5	Sep 30 (holiday) Sep 28 - Oct 2	OSv1: 16.5, 16.6, 17.3, 17.4 EN4, EN5	Sound intensity and sound intensity level (dB); sound field around point and line sources; inverse square law; acoustical attenuation; reflections at boundaries between two media; boundary conditions; amplitude reflection and transmission coefficients at a junction; energy reflection and transmission coefficients.		Lab 2 (except Wed sections)
Week 6	Oct 5 - Oct 9	OSv1: 16.5,16.6, 17.4, 17.5 EN6	Standing waves on a stretched string of fixed length: normal modes; acoustical standing waves: vibrations of air columns, normal modes; standing wave ratio; resonance.	Lab 2	
Week 7	Oct 12 (holiday) Oct 13 - Oct 16	OSv1: 16.5, 17.6, 17.7 OSv3: 3.1 EN7	Doppler effect; superposition of two harmonic waves of different frequencies: beats; coherent and incoherent sources; two-slit interference.		Lab 2 (Wed sections)
Week 8	Oct 19 - Oct 23	Midterm Exam: Wed. Oct. 21, 7:00-8:30 PM OSv3: 6.1, 6.2, 6.3, 7.1	Thin film interference; multiple reflections in a thin film; diffraction pattern of a single slit; circular aperture; resolving power.	Lab 3	
Week 9	Oct 26 - Oct 30	OSv3: 1.1, 1.2, 1.3, 1.6, 2.2	Emission and absorption of light; photoelectric effect		Lab 3
Week 10	Nov 2 - Nov 6	OSv3: 6.5	Photon momentum and probability waves; De Broglie waves; electron diffraction	Lab 4	
Week 11	Nov 9 - Nov 13	Term Break - No classes		No Labs	
Week 12	Nov 16 - Nov 20	OSv3: 7.2, 7.3, 7.6	Schrodinger's equation; Heisenberg's uncertainty principle; energy and wave functions of a trapped electron.		Lab 4
Week 13	Nov 23 - Nov 27	OSv3: 6.4, 7.5, 7.6	An electron in a potential well; the hydrogen atom: Bohr model.	Lab 5	
Week 14	Nov 30 - Dec 4	OSv3: 6.2, 6.4, 7.3	The hydrogen atom: Schrodinger's equation; electrical properties of metals.		Lab 5